23AID201: MODELING, SIMULATION & ANALYSIS

PROJECT TITLE: SMART VACUUM CLEANER

TEAM: A8

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INTRODUCTION

- Smart home appliances are popular for their ability to simplify daily tasks and improve effectiveness and it automates floor cleaning, reducing manual effort.
- Vacuuming alone may not suffice for maintaining floor hygiene in high dust or liquid spill areas and Incorporating mop feature into smart vacuum cleaner enhances its cleaning capabilities.
- Project focuses on creating smart vacuum cleaner and System aims to automate vacuuming and providing comprehensive cleaning solution.
- Integration of sensors, motor control, enables intelligent navigation.

LITERATURE REVIEW

Serial NO.	Titles	Key Contribution
1	A smart autonomous floor cleaner with an Androidbased controller	design and implementation of a smart autonomous floor cleaner with an Android-based controller
2	Low Cost Autonomous Robot Cleaner using Mapping Algorithm based on Internet of Things (IoT)	design and develop a low-cost Arduino autonomous robot cleaner using mapping algorithm to clean rooms.
3	Design and implementation of a cost effective vacuum cleaner robot	Mechanical, Electrical and software design of the smart vacuum cleaner
4	Designing a Smart Vacuum Cleaner	as a prototype, a robot that can navigate autonomously in its environment has been designed and implemented.
5	Robot Vacuum Cleaner Using SLAM and ROS	enhancing the autonomous capabilities of robot vacuum cleaners by integrating the Robot Operating System (ROS) and Simultaneous Localization and Mapping (SLAM) techniques.

RESEARCH GAP

Enhanced Virtual Sensor Integration for Improved Obstacle Avoidance:

- Although sensors like LiDAR and ultrasonic are used, there's limited research on their effectiveness in virtual environments for smart vacuum cleaners, specifically in handling diverse obstacles and adapting to complex room layouts autonomously.

Optimized Path-Planning Algorithms for Full Coverage:

- Existing algorithms in simulations are often not optimized for complete coverage without redundancy, which results in inefficiencies in cleaning. The research lacks effective path-planning solutions tailored for smart vacuum cleaners in simulated environments.

OBJECTIVES

Develop a Fully Autonomous Smart Vacuum Simulation in Webots:

- Create a virtual smart vacuum cleaner with autonomous navigation, equipped to identify and clean specific areas with detected dirt

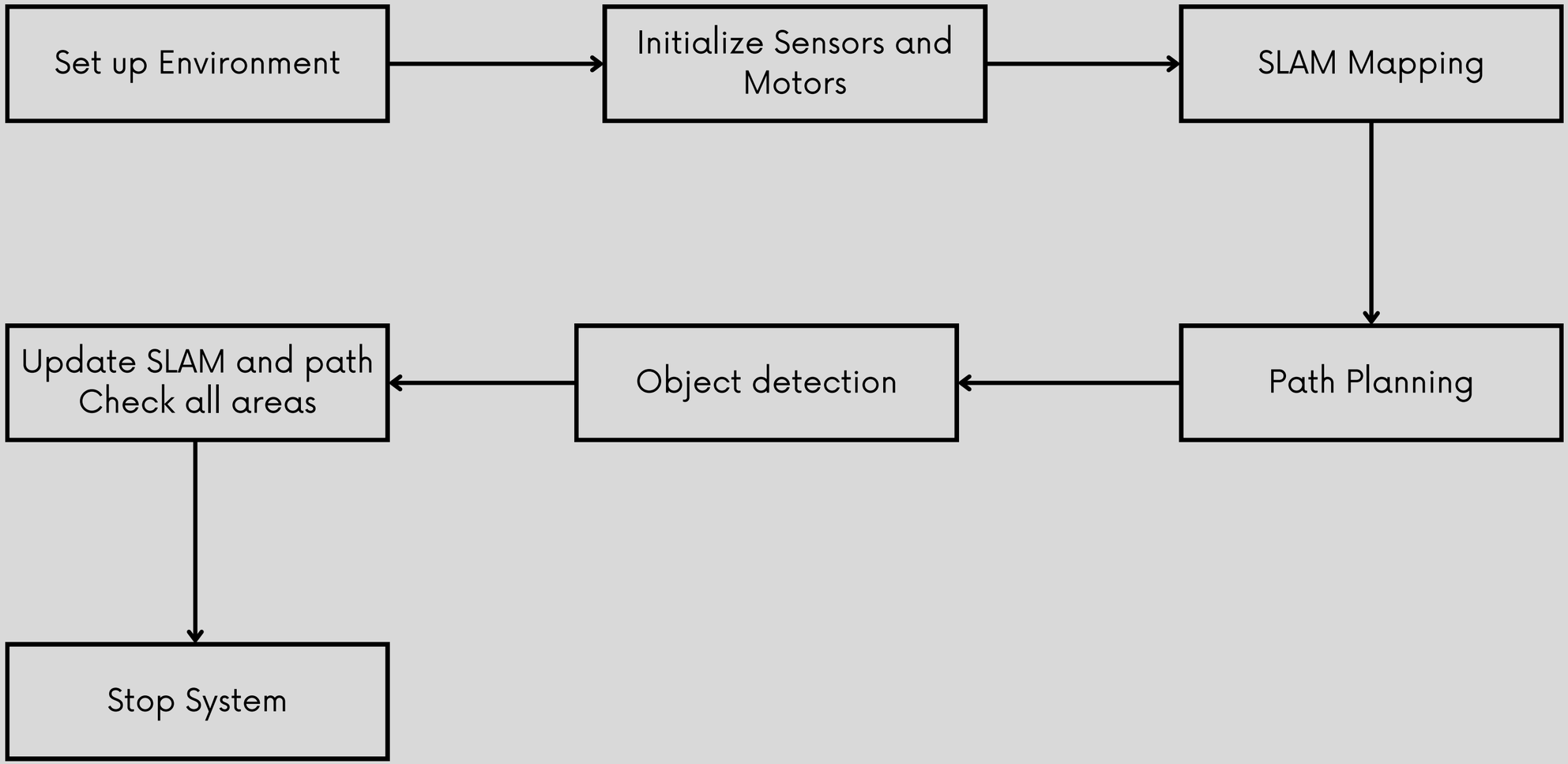
Implement Advanced Obstacle Detection and Avoidance Using Virtual Sensors:

- Integrate virtual sensors such as LiDAR and ultrasonic sensors to detect obstacles, enabling efficient navigation and maneuvering through complex room layouts without collisions.

Ensure Complete Room Coverage with Efficient Path Planning:

- Design path-planning algorithms that ensure comprehensive room coverage, minimizing missed spots, and adapting to varying room structures and obstacles.

Software


Set up Environment
Initialize Sensors and Motors
SLAM Mapping
Path Planning
Object detection
Update SLAM and path Check all areas
Stop System

METHODOLOGY

SLAM (Simultaneous Localization and Mapping): It enables a robot to map an environment and track its location simultaneously, essential for autonomous navigation. It combines localization (estimating position) and mapping (building an environment map) using data from sensors like Lidar and cameras. Challenges include handling sensor noise, data association, and real-time processing.

RESULTS

Final Result

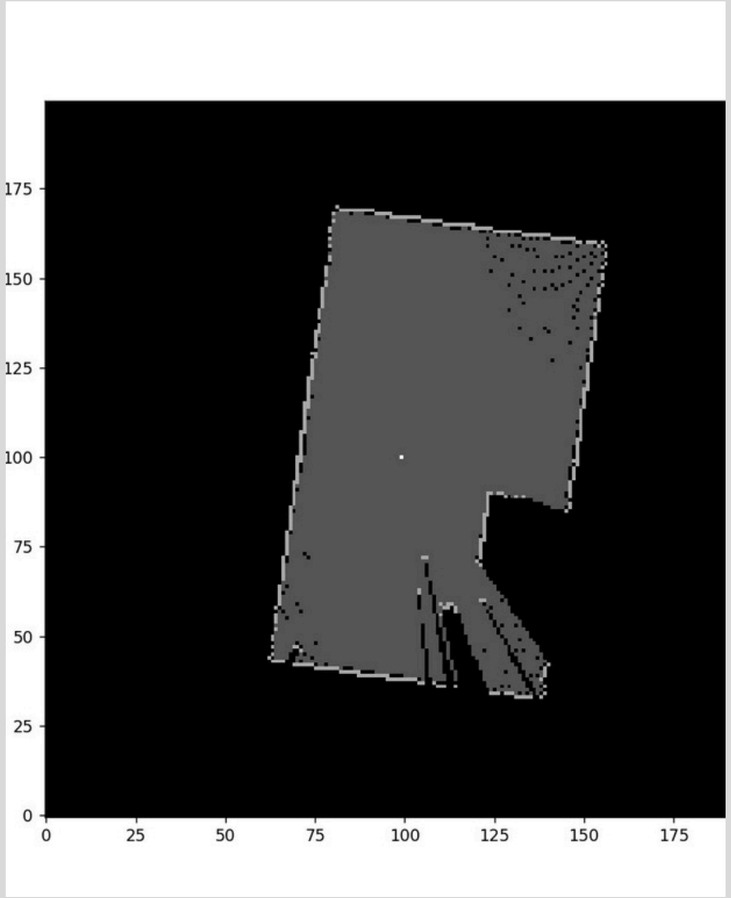


Fig 1

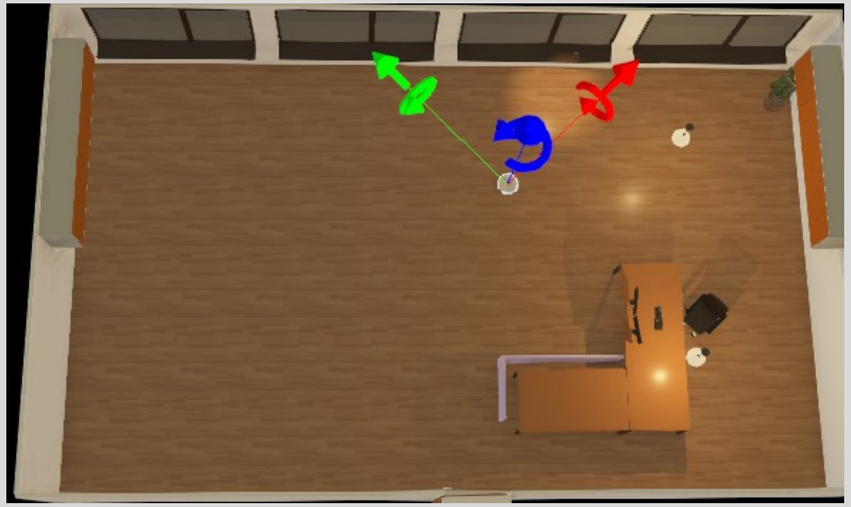


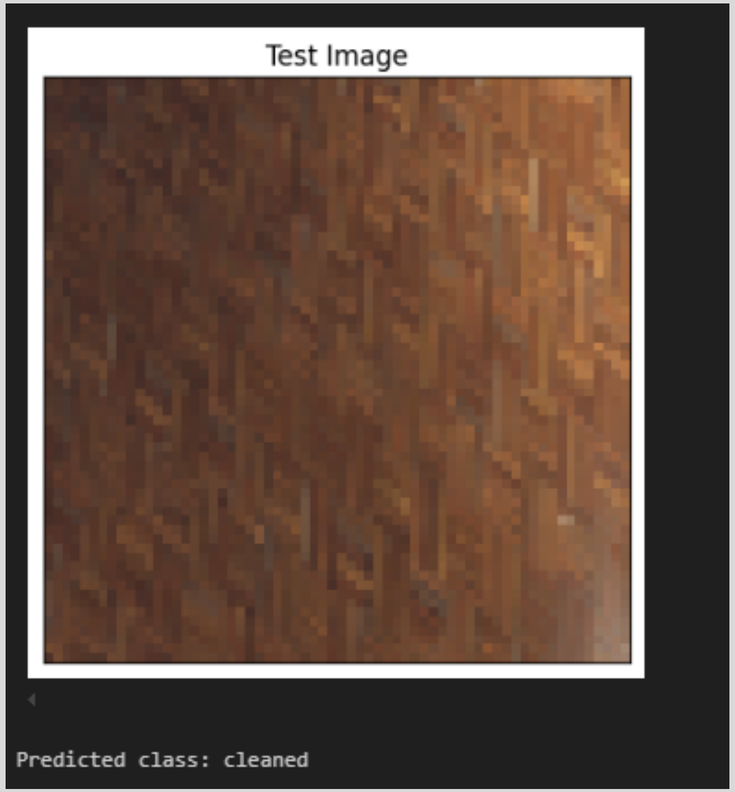
Fig 2

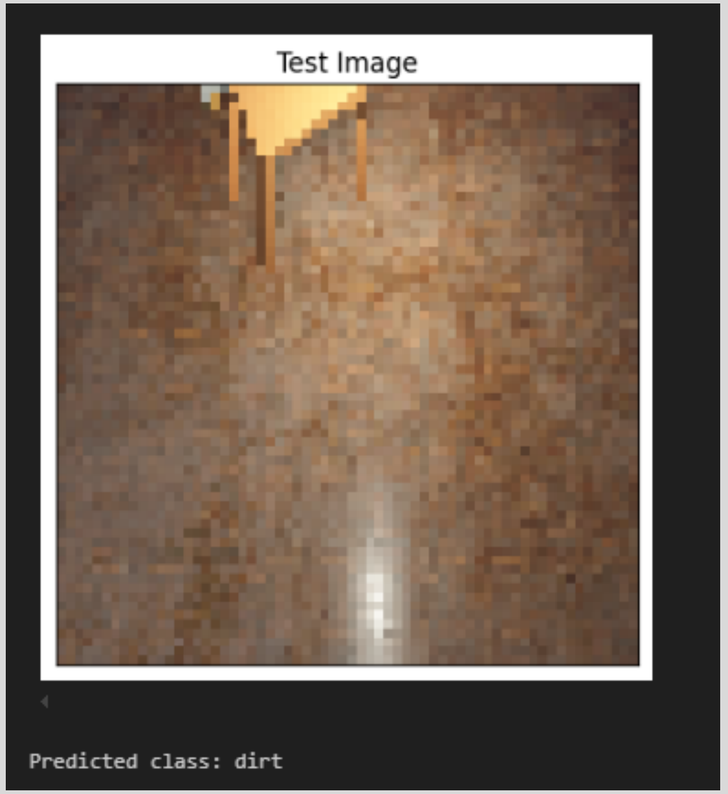


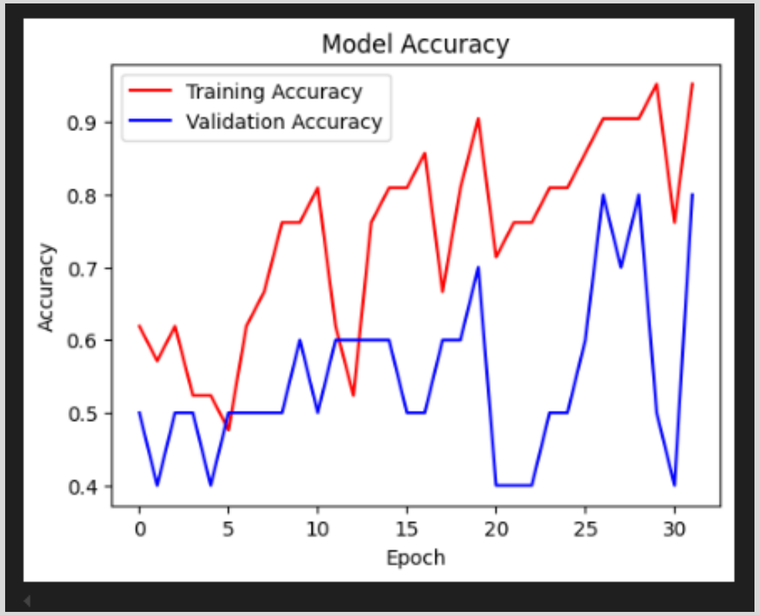
Fig 3

RESULTS

CNN(for Dirt detection)



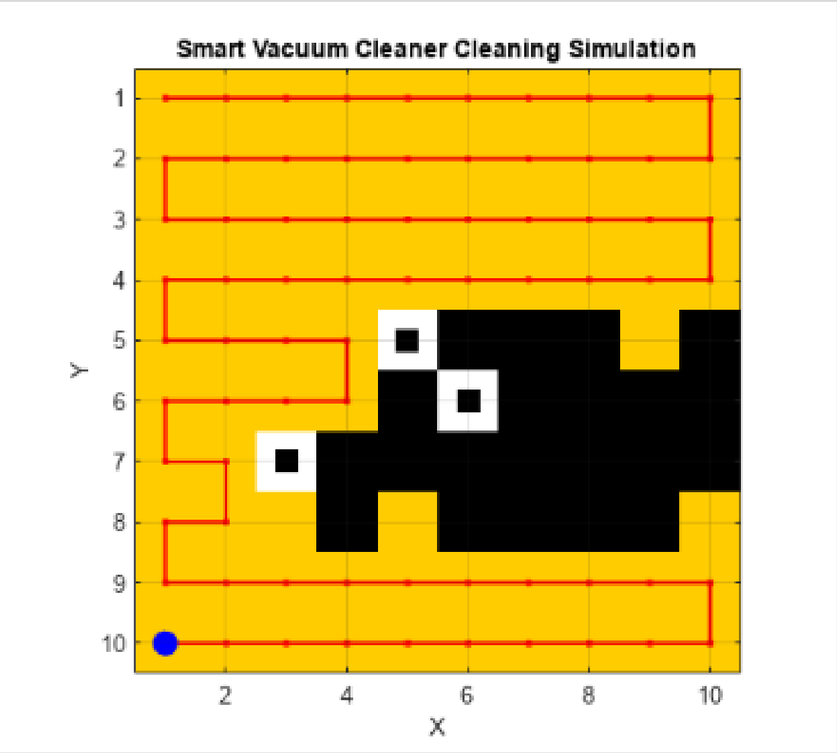




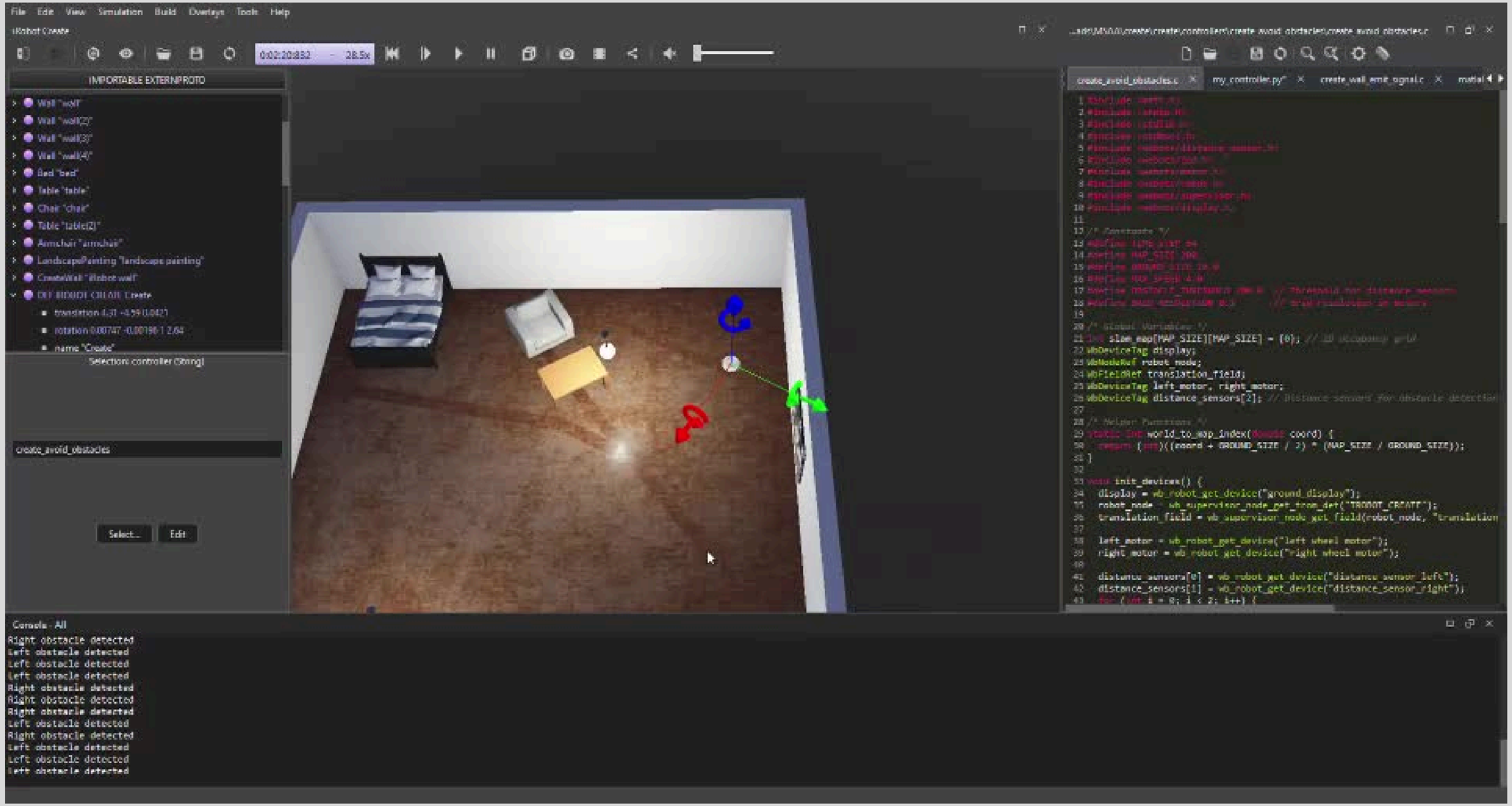

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 162ms/step - accuracy: 0.8571 - loss: 0.3413
Training accuracy: 0.8571428656578064
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 137ms/step - accuracy: 0.8000 - loss: 0.6181
Validation accuracy: 0.800000011920929
```

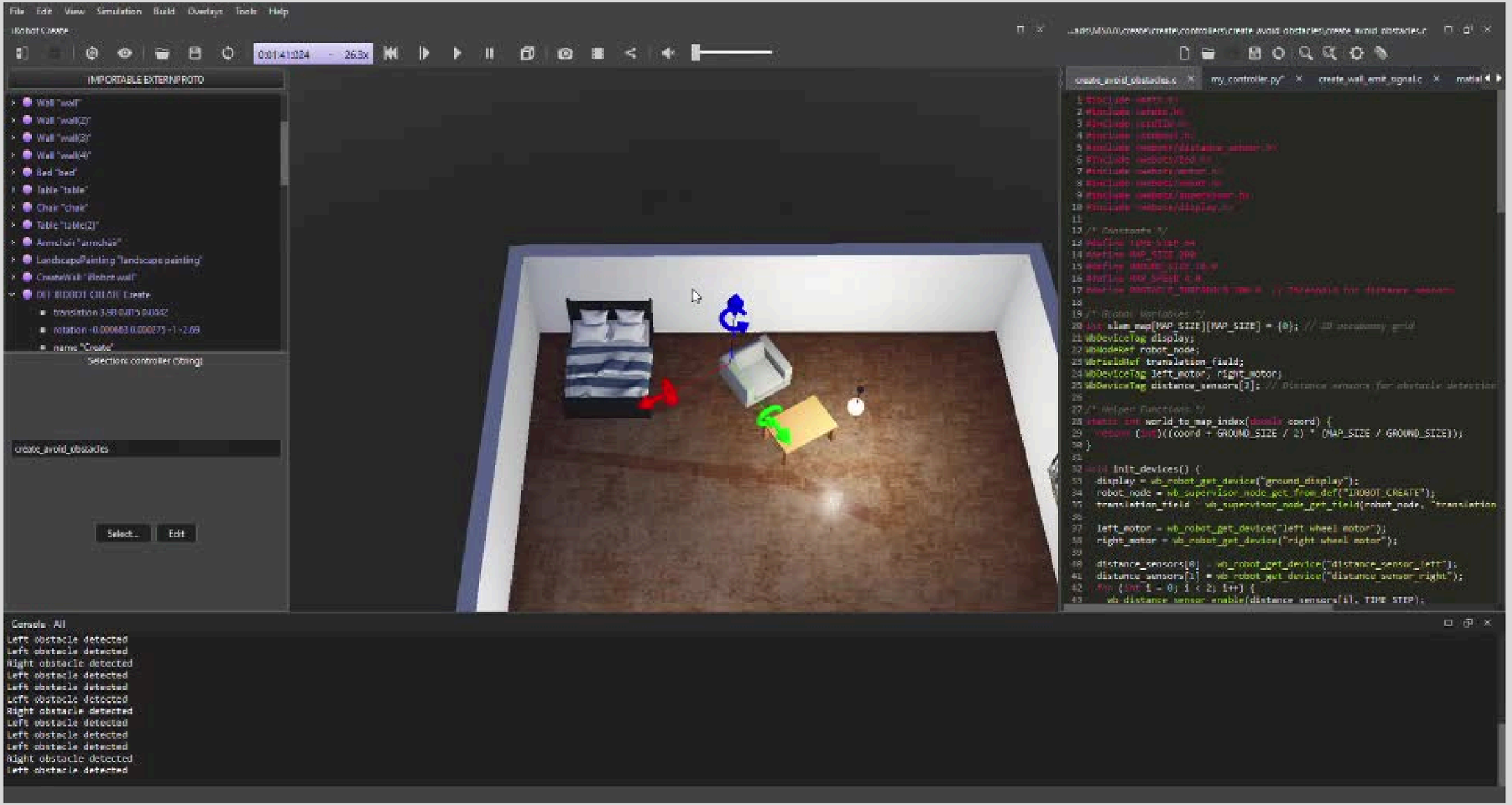
Matlab


Smart Vacuum Cleaner Cleaning Simulation
1
2
3
4
5
6
7
8
9
10
Y
2
4
6
8
10
X

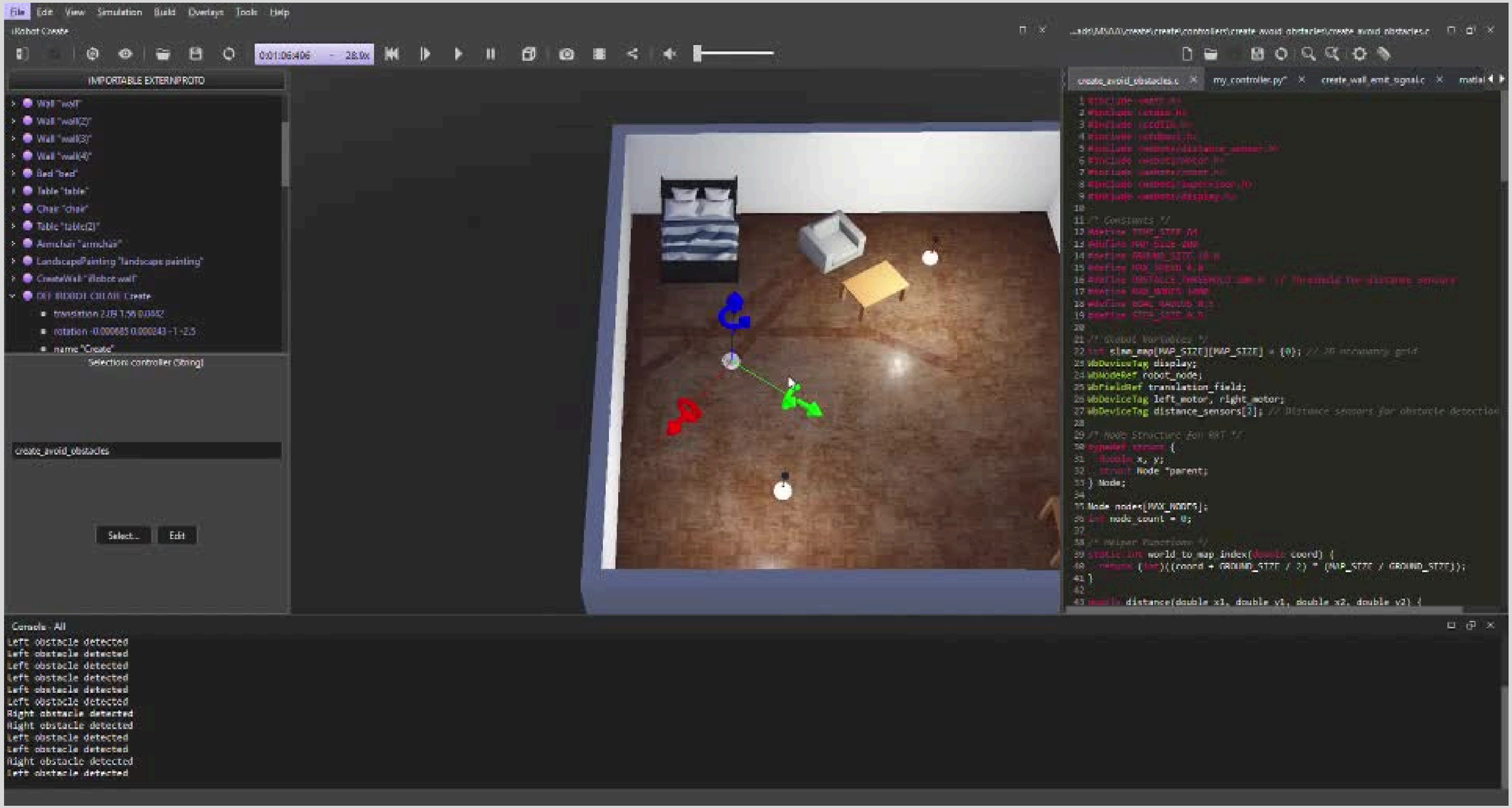
Boustreophedon



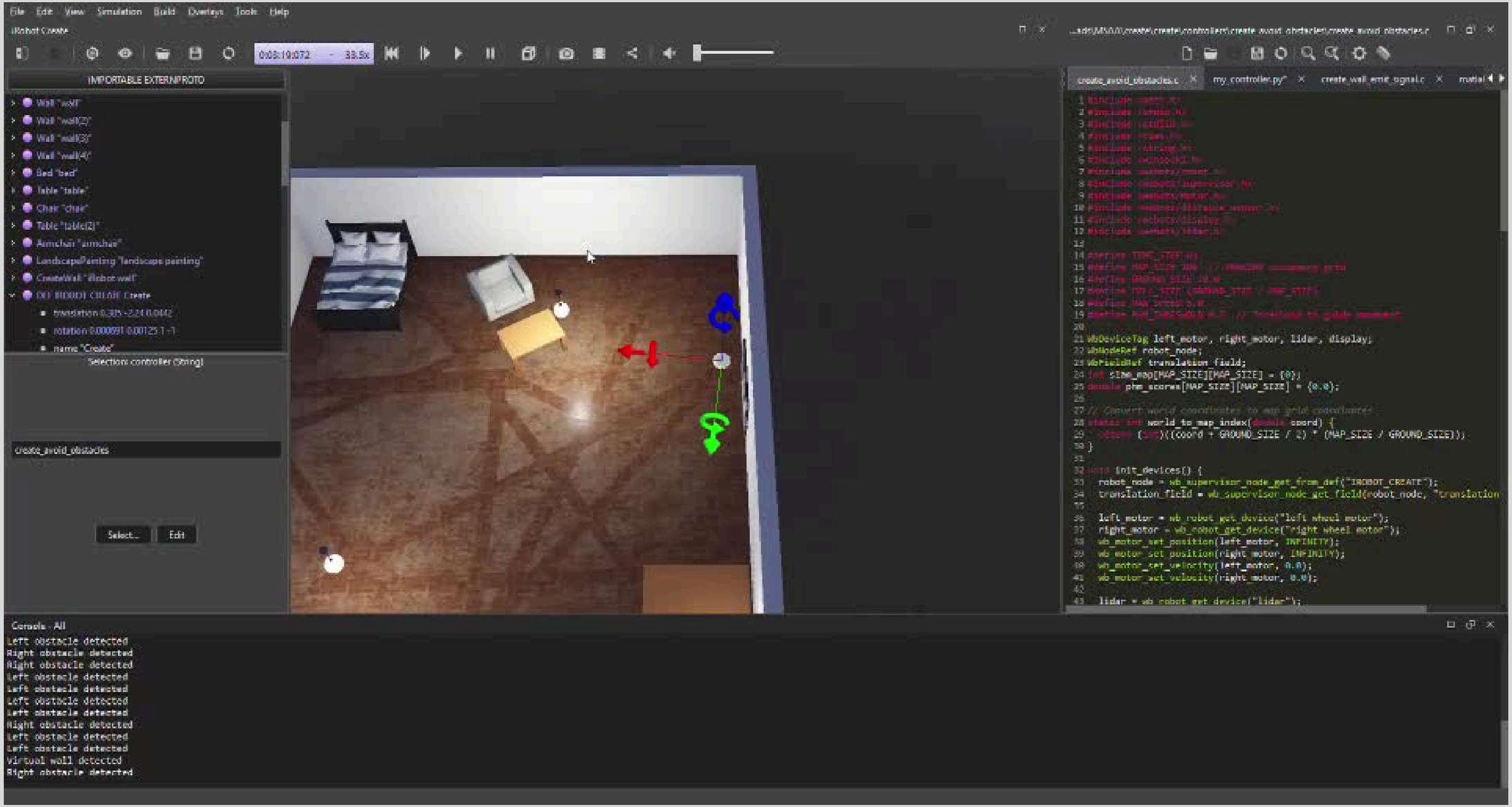
Wave front


File Edit View Simulation Build Overlays Tools Help
create_avoid_obstacles
Console - All
Left obstacle detected
Left obstacle detected
Right obstacle detected
Left obstacle detected
Left obstacle detected
Left obstacle detected
Right obstacle detected
Left obstacle detected
Left obstacle detected
Left obstacle detected
Right obstacle detected
Left obstacle detected

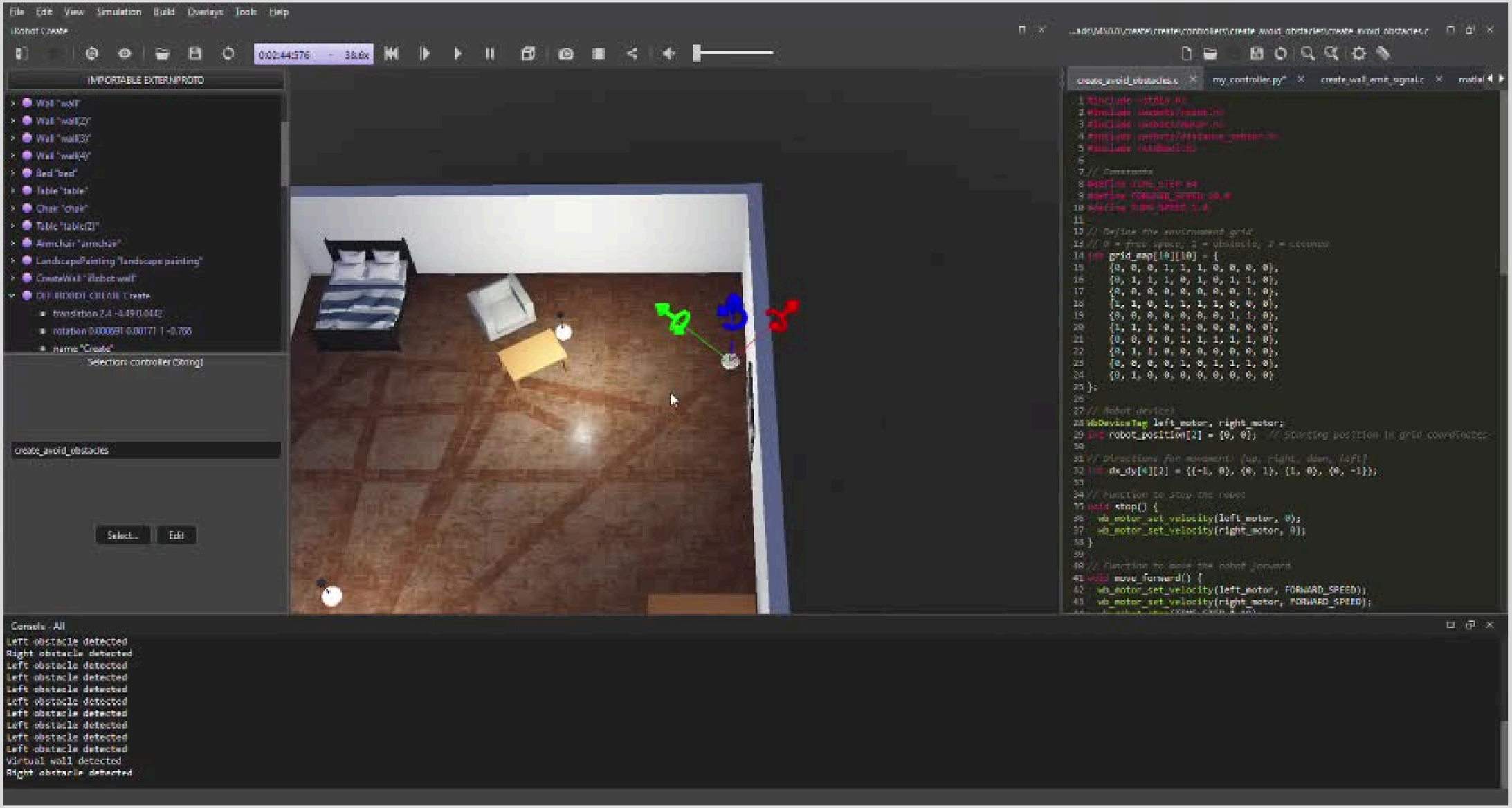
Zig-Zag


File Edit View Simulation Build Overlays Tools Help
IMPORTABLE EXTERNPROTO
Wall "wall"
Wall "wall(2)"
Wall "wall(3)"
Wall "wall(4)"
Bed "bed"
Table "table"
Chair "chair"
Table "table(2)"
Armchair "armchair"
LandscapePainting "landscape painting"
CreateWall "Robot wall"
DEF ROBOT CHILDS Create
translation 2.09 1.58 0.082
rotation -0.000683 0.000243 -1 -2.5
name "Create"
Selection: controller (String)
create_avoid_obstacles
Select...
Edit
create_avoid_obstacles.c
my_controller.py*
create_wall_emit_signal.c
Console - All
Left obstacle detected
Left obstacle detected
Left obstacle detected
Left obstacle detected
Left obstacle detected
Left obstacle detected
Right obstacle detected
Right obstacle detected
Left obstacle detected
Left obstacle detected
Right obstacle detected
Left obstacle detected

PHM(Probabilistic Harmonic Mean)



Breadth-First Search



CONCLUSION

Implemented slam and path planning algorithms like A*, Dijkstra's, boustrophedon, Zig-Zag, Spiral.

REFERENCES

1. H. A. S. Hossen Prayash, M. R. Shaharear, M. F. Islam, S. Islam, N. Hossain, and S. Datta, "Designing and Optimization of An Autonomous Vacuum Floor Cleaning Robot," in 2019 IEEE International Conference on Robotics, Automation, Artificial-intelligence and Internet-of-Things (RAAICON), Dhaka, Bangladesh, Nov. 2019, pp. 1-7. doi: 10.1109/RAAICON48939.2019.11.

2. M. Arjun, C. A. Oliver, and A. M. Joseph, "Robot Vacuum Cleaner Using SLAM and ROS," 2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS), Chikkaballapur, India, 2024, pp. 1-5. doi: 10.1109/ICKECS61492.2024.10616915. Keywords: Productivity; Simultaneous localization and mapping; Vacuum systems; Service robots; Motors; Cleaning; Sensors; Robot vacuum cleaner; Robot Operating System; Simultaneous Localization and Mapping.

3. P. Vibhandik and Z. Khan, "Autonomous Vacuum Cleaning Robot Using Sonar Navigation and SLAM," International Research Journal of Engineering and Technology (IRJET), vol. 8, no. 7, pp. 1905-1910, July 2021